Lecture 2

Drug, target and disease databases

AI- & data-driven drug design – An introduction

A **database** is a digital repository for storing, managing and securing organized collections of data

1. Why do databases matter

Databases are essential for research: they give us access to far more information than any of us could read or generate ourselves.



Volume

Research produces far more data than anyone can follow; databases keep track of what is known.



Standardization

Results from thousands of labs share common formats and identifiers, so they can be compared.



Access and reuse

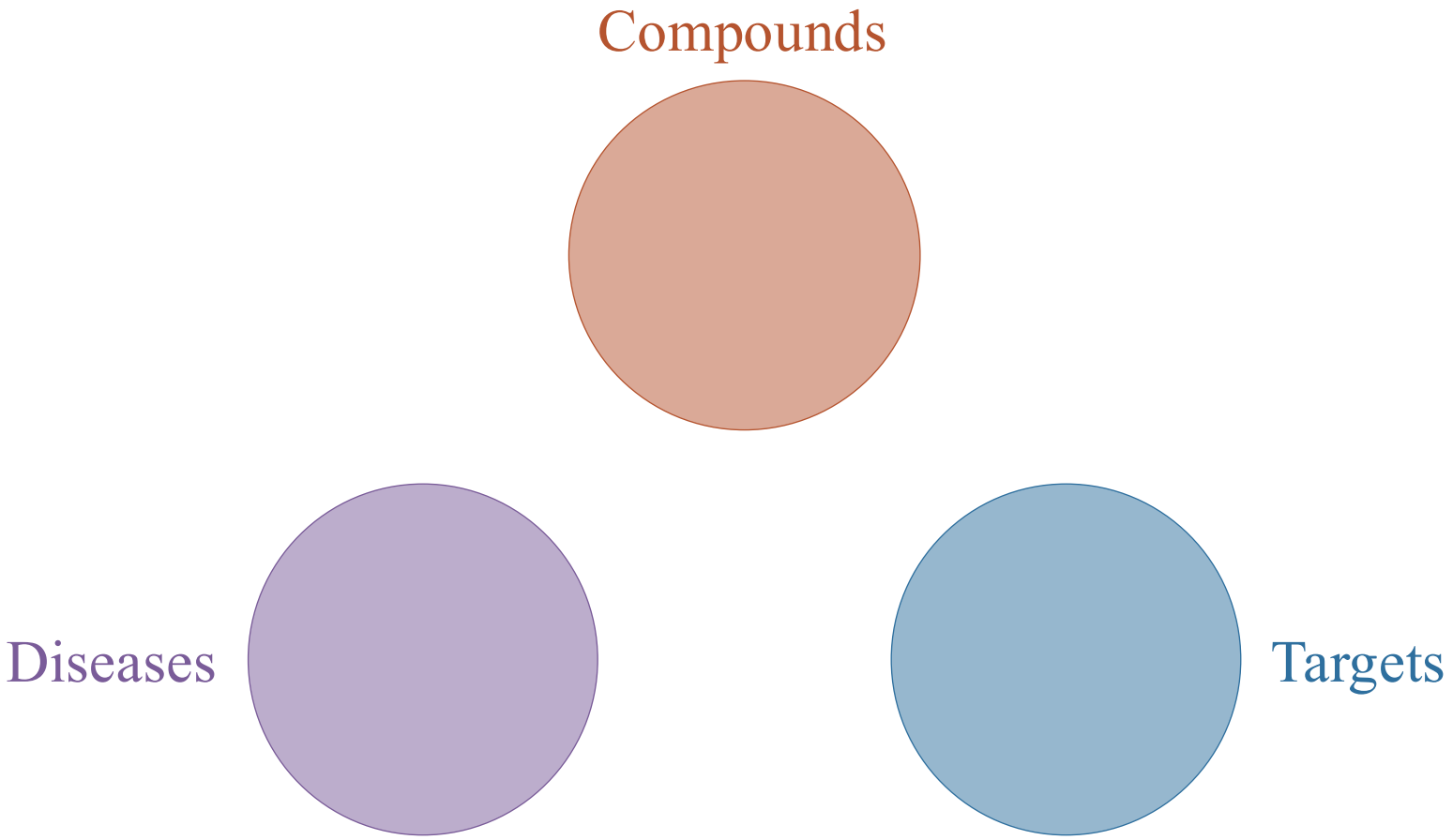
Data can be queried, re-analysed and reused for questions the authors never asked.



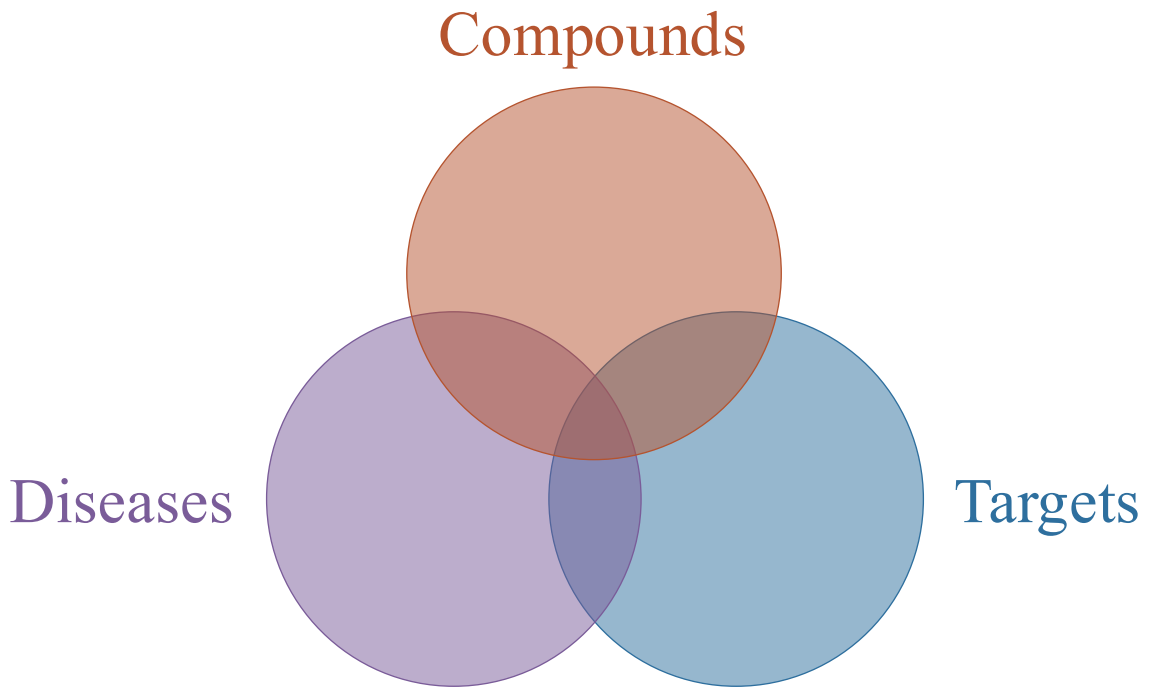
Integration

Shared identifiers link databases to answer questions that no single study could.

1. Why do databases matter in drug design



1. Why do databases matter in drug design



2. Types of databases

	Compound-centric	Target-centric	Disease-centric	Association
Definition	Molecules	Protein and genes	Diseases and phenotypes	Links between entities
Data	Chemical structures Physicochemical properties Bioactivities	Sequence 3D structure Function Known ligands Expression profile	Genetic associations Pathways Phenotypes	Compound–target activities Target–disease evidence Drug–indications
Examples	PubChem, ChEMBL, ZINC	UniProt, PDB, AlphaFold DB	OMIM, Orphanet, ClinVar, GWAS Catalog	OpenTargets, Pharos

2. Types of databases

Besides their main focus, databases differ in:

Curation

More curation: fewer but more reliable records.

- Expert-curated
- Extracted from literature
- Deposited by authors
- Text-mined

Data type

Training on predicted data feeds another model's assumptions into yours.

- Measured
- Predicted

Access & license

Can the data be used or redistributed commercially?

- Open
- Free for academics
- Commercial

Updates

Versioned releases make results reproducible.

- Continuous
- Versioned releases

Database selection criteria

1. Which entity is my question about:
compound, target, disease, or a link between them?
2. Do I need measured data, or are predictions acceptable?
3. Do I need quantity (for training) or quality (for validation or decisions)?
4. Does the license allow what I plan to do with the data?

2.1. Drug databases



~3 million compounds, 20 million+ activity values

Manually curated database of bioactive molecules with drug-like properties.



100 million+ compounds

Largest collection of freely accessible chemical information, with chemical and physical properties, and some biological activities, safety and toxicity and more.



200.000 naturally occurring/synthetic compounds

Manually curated database and ontology of chemical entities (mostly small molecules).



>100 Billions of molecules

Largest commercially available, synthetically accessible chemical library designed for virtual screening, hit finding and drug discovery.

2.1. Drug databases – Compound identifiers

SMILES

Simplified Molecular Input Line Entry System

For writing and modelling: atoms, bonds, branches and rings in one short line.

```
C[C@H](N)C(=O)O
```

+ readable, ML input

– many SMILES per molecule

InChI

IUPAC International Chemical Identifier

For identifying: one open algorithm, built in layers (formula, connectivity, H, charge, stereo).

```
InChI=1S/C3H7NO2/c1-
2(4)3(5)6/h2H,4H2,1H3,(H,5,
6)/t2-/m0/s1
```

+ unique across software

– long, hard to read

InChIKey

27-character hash of the InChI

For searching and matching: skeleton (14) · stereo and other layers (10) · protonation (1).

```
QNAYBMKLOCPYGJ-REOHCLBHSA-N
```

+ fixed length, ideal database key

– cannot be turned back into a structure

All three examples encode L-alanine.

2.1. Drug databases – Compound identifiers

Molecule	SMILES	InChIKey	What it shows
Ethanol	CCO OCC C(O)C	LFQSCWFLJHTTHZ-UHFFFAOYSA-N	Three valid SMILES, one InChIKey
L-alanine	C[C@H](N)C(=O)O	QNAYBMKLOCPYGJ-REOHCLBHSA-N	Stereochemistry (@ / @@) changes only the second block of the InChIKey
D-alanine	C[C@@H](N)C(=O)O	QNAYBMKLOCPYGJ-UWTATZPHSA-N	
Alanine, stereo undefined	CC(N)C(=O)O	QNAYBMKLOCPYGJ-UHFFFAOYSA-N	UHFFFAOYSA = no stereo information
2-Hydroxypyridine **2-Pyridone**	Oc1ccccn1 O=c1cccc[nH]1	UBQKCCHYAOITMY-UHFFFAOYSA-N	Two tautomers: different SMILES, same InChI and InChIKey
Acetic acid **Sodium acetate**	CC(=O)O CC(=O)[O-].[Na+]	QTBSBXVTEAMEQO-UHFFFAOYSA-N VMHLLURERBWHNL-UHFFFAOYSA-M	A salt gets its own key (M = proton removed), so standardise before comparing

2.2. Target databases



~ 500.000 reviewed and 149M unreviewed proteins

World's leading high-quality, comprehensive and freely accessible resource of protein sequence and functional information.



>250.000 Experimental 3D structures

Structures from X-ray crystallography, cryo-EM and NMR



>200M Predicted 3D structures

Models for most known proteins; check the confidence (pLDDT)



Tissue and cell expression of proteins

Human Protein Atlas, HPA: Protein expression across tissues and cells

2.2. Target databases – Family specific

IUPHAR/BPS

~ 3k targets, ~10k molecules, ~2k approved drugs

An expert-curated resource of pharmacological targets and drugs and tool compounds



810 GPCRs, 2.5k structures, 250k ligands

GPCRdb: Reference data, analysis, visualization, experiment design and data deposition.

KLIFS

337 Kinases, 7k structures, 4k ligands

All kinase-ligand structures from the PDB with interaction fingerprints

2.3. Disease databases



Disease ontology

Cross-referenced to HPO, OMIM, Orphanet and others



Experimental factor ontology

Systematic, standardized description of experimental variables including diseases



Human Phenotype Ontology

Standardized, controlled vocabulary used to describe phenotypic abnormalities



International Classification of Diseases

Global standard for diagnostic health information.

2.4. Integrated resources

Open Targets

~ 79k targets, ~46k diseases, ~19k clinical compounds

Comprehensive tool that supports systematic identification and prioritization of potential therapeutic drug targets.

PHAROS

~4.5 k targets across GPCRs, ion channels and kinases

Understanding of understudied protein families through the development of an information repository and new tools.

TTD

~4k targets and ~40k clinical compounds

Therapeutic Targets Database, TTD: Known and explored therapeutic protein and nucleic acid targets, targeted disease, pathway information and the corresponding drugs.

3. Data quality and integration

The most common problems in database records:

Assay heterogeneity

The same compound–target pair often differs by ~0.5 log units between labs, limiting model accuracy.

Mixed activity types and units

IC50, Ki, Kd and % inhibition are not directly comparable; some values are censored (“>10 µM”).

Publication and selection bias

Mostly actives on popular targets; inactive compounds and understudied targets are underrepresented.

Wrong or ambiguous targets

Assays on cells, other species or protein complexes assigned to a single human protein.

3. Data quality and integration

Good practice

Record database version and access date

e.g. ChEMBL 37, accessed on 9 November 2026

Cite the latest database publication

If using multiple database resources, cite all of their publications.

Filter by confidence

ChEMBL confidence score; UniProt reviewed entries

Prefer expert-curated databases

when quality matters more than quantity

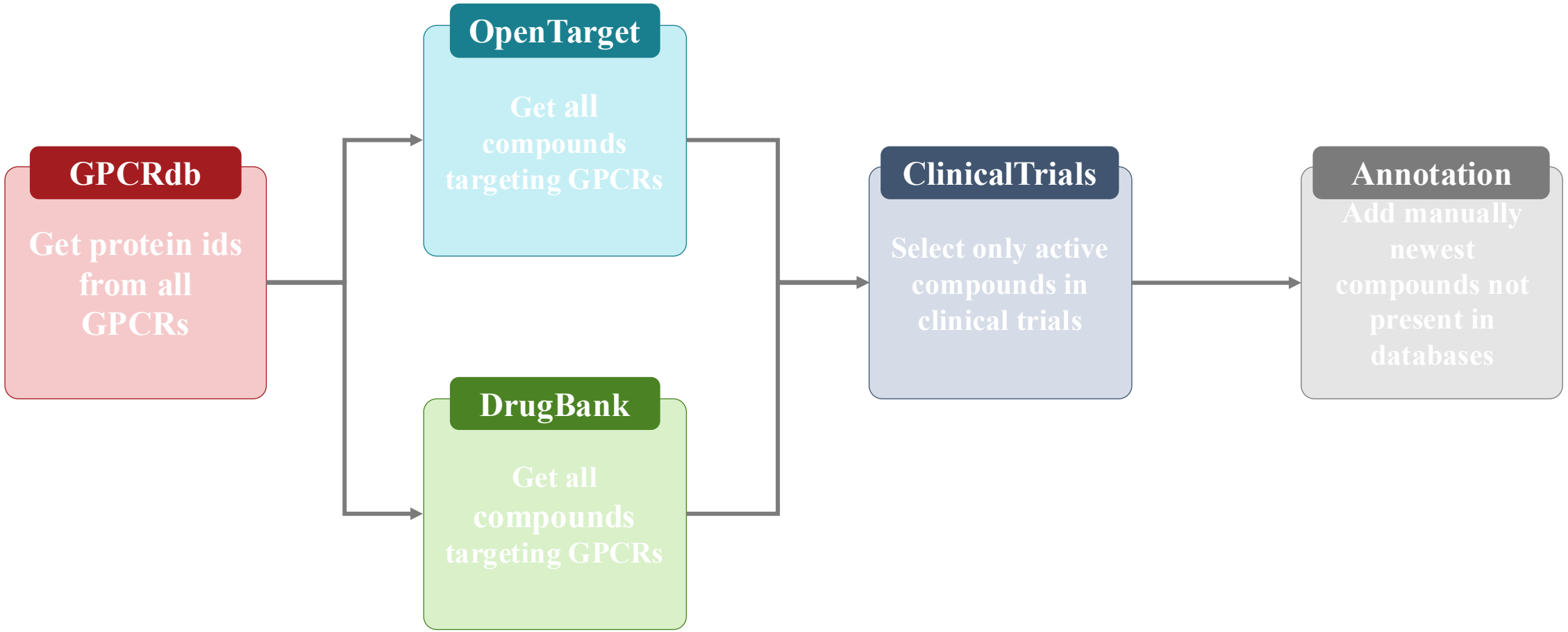
Standardise structures and aggregate replicates

e.g. merge salts of one compound; median per compound

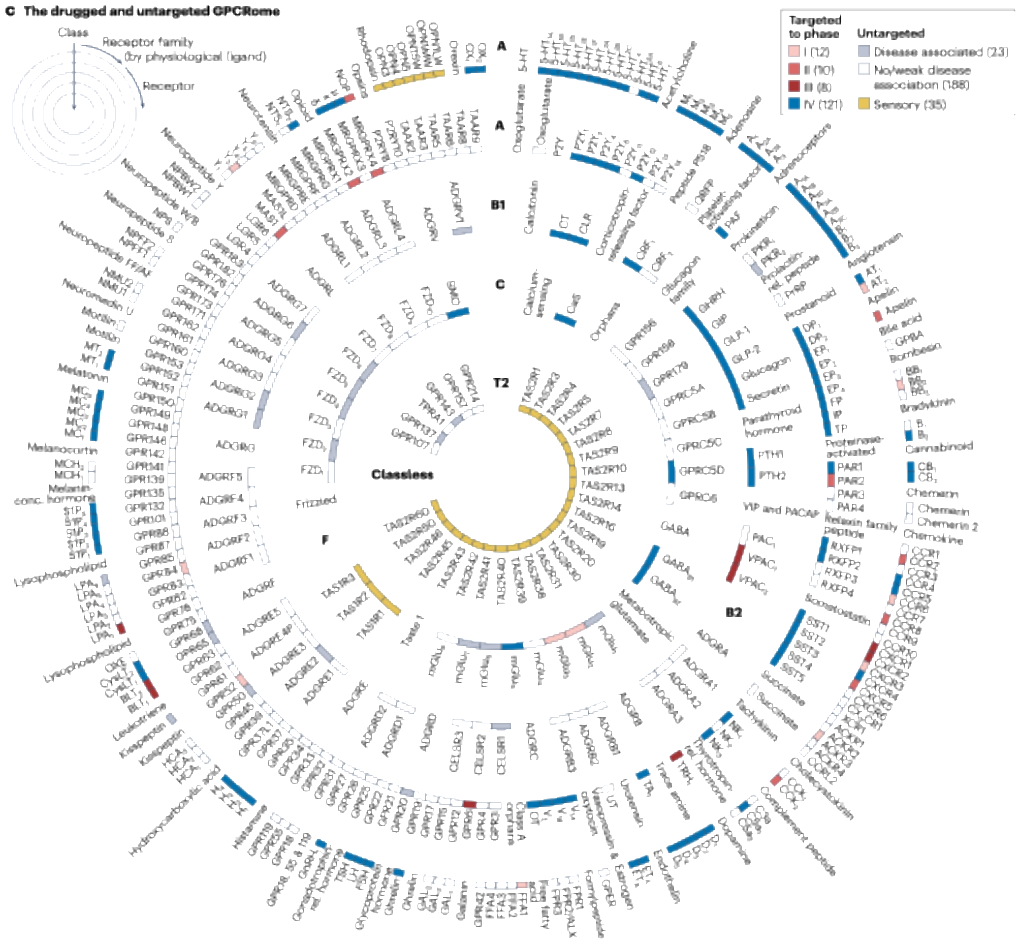
Check licenses before redistribution

open, academic-only or commercial

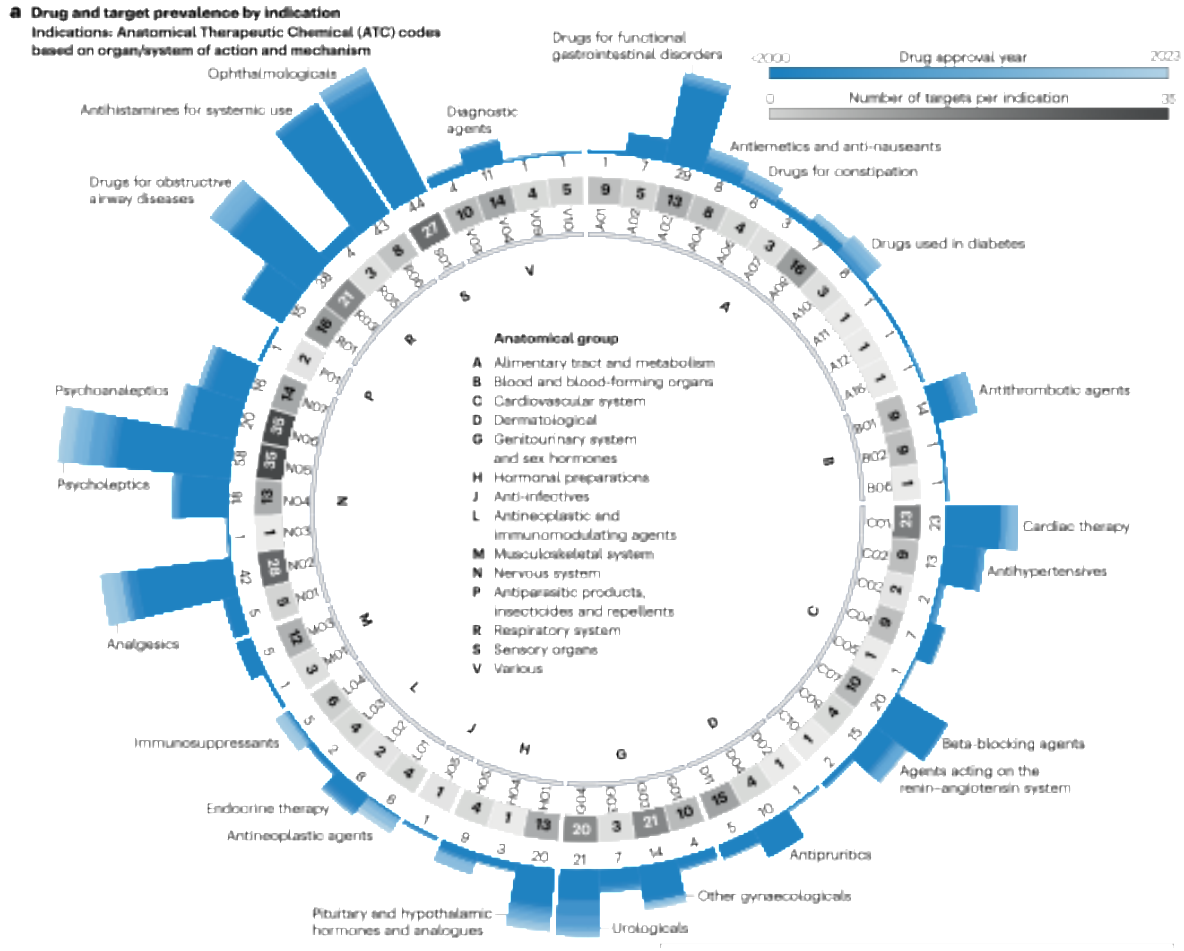
4. Example: GPCR drug discovery



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5. Programmatic access

Task: Find all human D2 receptor Ki values (ChEMBL)

1. **Find the target**
   human dopamine D2 receptor
2. **Retrieve Ki values**
   binding assays with a pChEMBL value
3. **Clean**
   keep exact values (“=”), median per compound
4. **Save**
   record the ChEMBL release and filters

5. Programmatic access

Task: Find all human D2 receptor Ki values (ChEMBL)

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```python
# 1. Find the target: human dopamine D2 receptor
target = new_client.target.filter(
    pref_name__iexact="Dopamine D2 receptor",
    organism="Homo sapiens",
    target_type="SINGLE PROTEIN",
).only(["target_chembl_id", "pref_name"])[0]
target_id = target["target_chembl_id"]
print(target_id, target["pref_name"])

# 2. Retrieve binding-assay (B) Ki values that have a pChEMBL value
acts = new_client.activity.filter(
    target_chembl_id=target_id,
    standard_type="Ki",
    assay_type="B",
    pchembl_value__isnull=False,
).only(["molecule_chembl_id", "canonical_smiles", "standard_relation",
        "pchembl_value", "document_chembl_id"])
df = pd.DataFrame(acts)
print(len(df), "Ki records")

# 3. Clean: keep exact values ("="), then one median value per compound
df = df[df["standard_relation"] == "="].copy()
df["pchembl_value"] = df["pchembl_value"].astype(float)
table = (df.groupby(["molecule_chembl_id", "canonical_smiles"])["pchembl_value"]
           .agg(pchembl="median", n_measurements="size")
           .reset_index()
           .sort_values("pchembl", ascending=False))
print(len(table), "unique compounds")
```

Take-home messages

1. Databases are the foundation of AI- and data-driven drug design.
With increasing performance of AI models, accuracy depends on data quality.

2. Identifiers and ontologies make integration possible;
curation and filtering make it trustworthy

3. Family-specific resources and expert curation add depth to general databases

4. Scripting data access makes the work reproducible

Learning outcomes

- Understand why databases are essential for AI- and data-driven drug design

- Identify the most relevant databases for drug design, and classify them as compound-, target-, disease- or integrated resources

Questions & comments?